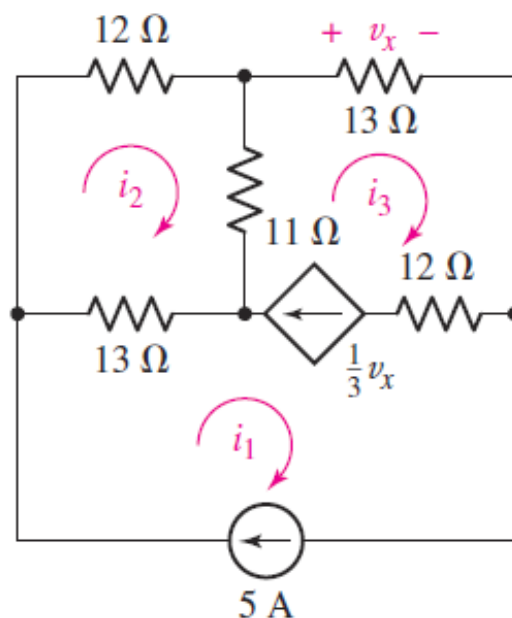


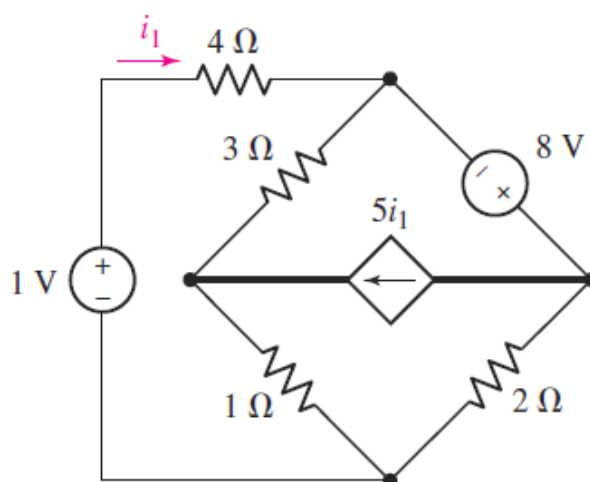
EC1101 – Tutorial Sheet 3

- 1 Find $\mathbf{i_1}$ to $\mathbf{i_3}$ using mesh analysis.



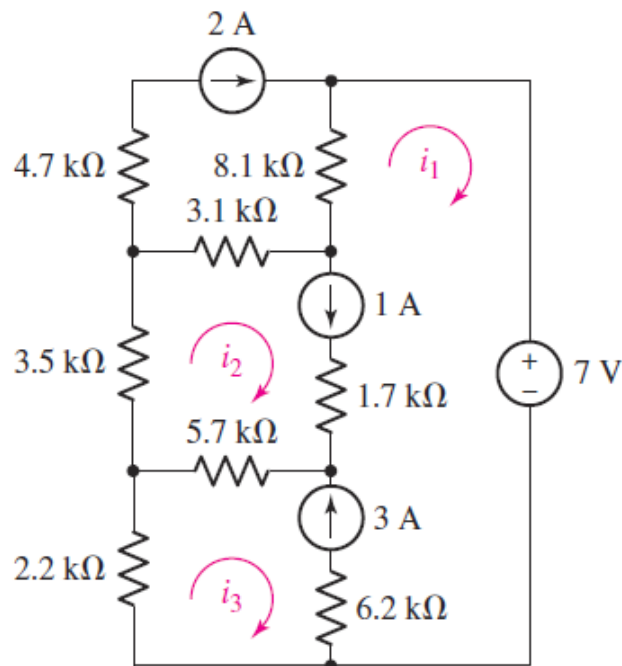
$$\mathbf{i_1 = 5\text{ A}, i_2 = 1.347\text{ A}, i_3 = -1.5\text{ A}.$$

- 2 Find $\mathbf{i_1}$ and the power supplied by the 1V source using mesh analysis.



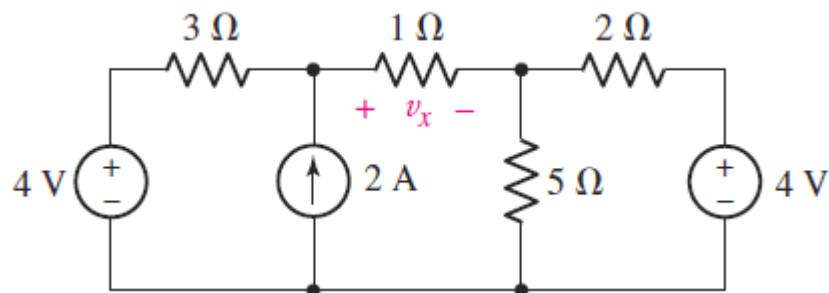
$$\mathbf{i_1 = 19\text{ A}; P = 19\text{ W}}$$

-
- 3 Calculate the 3 mesh currents labelled in the circuit below.



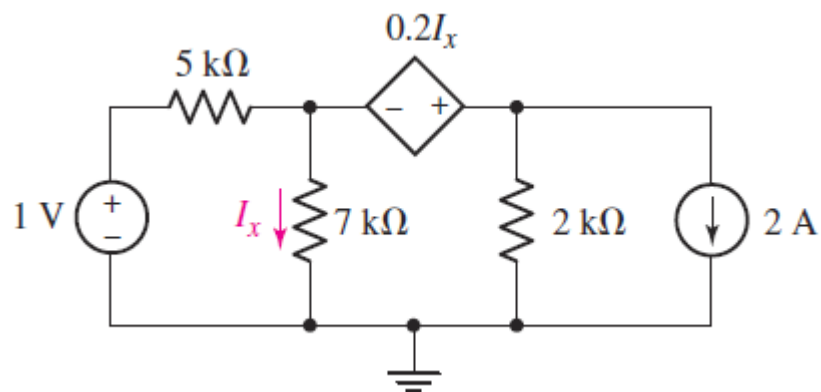
$$i_1 = 911 \text{ mA}, i_2 = 1.911 \text{ A} \text{ and } i_3 = -2.089 \text{ A}$$

-
- 4 Using superposition, determine the voltage v_x in the circuit.



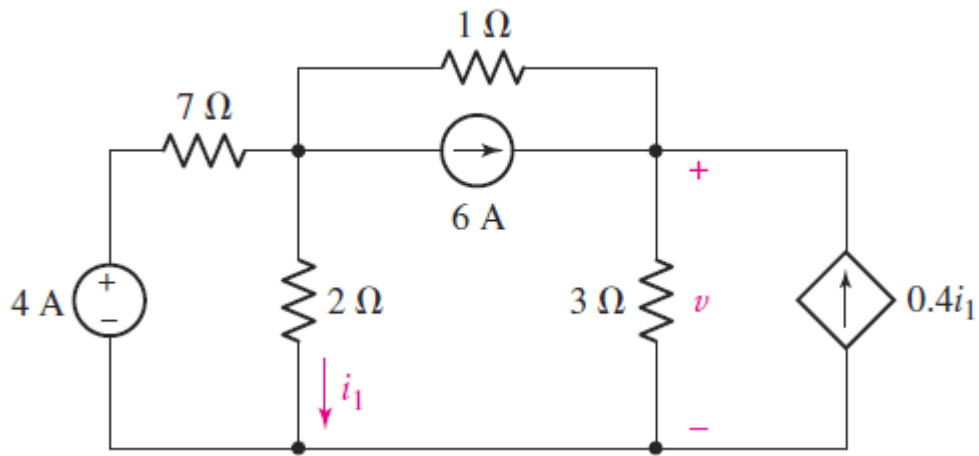
$$v_x = 1.316 \text{ V}$$

-
- 5 Determine the value of I_x in the circuit below using superposition.



$$I_x = 339 \text{ mA}$$

-
- 6 Employ superposition to determine the individual contribution from each independent source to the voltage v and current i_1 .



$v = 3.375 \text{ V}; i_1 = -625 \text{ mA}$
